

# DYNAMIC STATE SPACE MODELS FOR CROSS – MODALITY FUSION

Jinkai Yao<sup>\*,★</sup>      Jiaze Jin<sup>\*,★</sup>      Huiying Xu<sup>†,‡,★</sup>      Zeyu Wang<sup>★</sup>  
 Xinzhong Zhu<sup>‡,★,§</sup>      Hongbo Li<sup>§</sup>      Yue Hu<sup>★</sup>

<sup>‡</sup> Zhejiang Key Laboratory of Intelligent Education Technology and Application,  
 Zhejiang Normal University, Jinhua, China.

<sup>★</sup> School of Computer Science and Technology, Zhejiang Normal University,  
 Jinhua, China.

<sup>§</sup>Geekplus Technology

## ABSTRACT

This paper proposes a dynamic approach to optimizing State Space Models (SSMs) to address feature misalignment in cross-modality fusion. We design a novel modality fusion module based on dynamic alignment and state-space extraction, effectively mitigating modality discrepancies. Furthermore, we propose an innovative dynamic adaptive downsampling module, designed to enhance salient feature representation, with residual connections ensuring information integrity. Experimental results show that our method outperforms existing approaches, achieving 1.6% higher mAP than the second-best method, C<sup>2</sup>Former, on the LLVIP dataset. Our approach demonstrates improved performance and robustness, particularly under challenging conditions such as poor lighting and adverse weather. This work provides the first demonstration of dynamic SSMs for cross-modal fusion, establishing a new benchmark.

**Index Terms**— Cross-Modality Fusion, State Space Model, Dynamics Analysis, MSOD.

## 1. INTRODUCTION

The integration of infrared (IR) and RGB images in multispectral object detection (MSOD) leverages their complementary physical textures and thermal structures, enabling cross-modality fusion to mitigate noise and environmental challenges inherent in single modalities, thereby enhancing feature representation. Existing fusion approaches are predominantly based on Convolutional Neural Networks (CNNs) [1] or Vision Transformers (ViTs) [2]. CNNs achieve promising results through local receptive fields [3, 4], whereas ViTs leverage self-attention to model long-range dependencies [5, 6]; however, both approaches still encounter challenges such as modality discrepancy, computational overhead, and weak feature correlation.

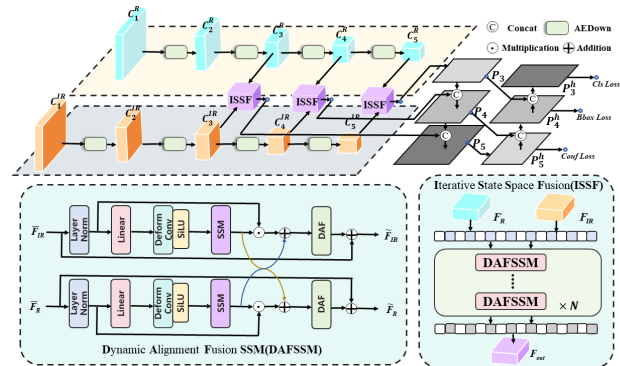
Recently, State Space Models (SSMs) [7–9] have been introduced into cross-modality tasks owing to their linear complexity and global receptive field. The SSM mechanism scans the input image in four directions—horizontally and vertically—aggregating features to capture complex spatial relationships and enable comprehensive contextual understanding, thus supporting global modeling.

<sup>\*</sup>Equal contribution

<sup>†</sup>Corresponding author.

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Nevertheless, their unidirectional modeling nature often overlooks local receptive fields, constrains adjacent pixel dependencies, and struggles with misaligned features in fusion.



**Fig. 1:** Architecture of our approach. The backbone consists of two components: a dual-stream multi-scale extraction network enhanced by adaptive feature importance, and three ISSF modules integrating multiple SSM-based dynamic feature alignment fusion (DAFSSM), applied at the  $\{C_3, C_4, C_5\}$  layers for cross-modality fusion. The resulting feature maps are input into the PAFPN, which generates  $\{P_3, P_4, P_5\}$  features, passed to the decoupled head for detection output. The design details of the DAFSSM and ISSF modules are shown at the bottom.

To overcome these drawbacks, we present a dynamic perspective for cross-modality fusion leveraging SSMs. Specifically, we introduce morphological adaptivity to dynamically enhance salient features before feeding them into the SSM, followed by dynamically generated convolution kernels to address local constraints and feature misalignment. Our main contributions are summarized as follows:

- We design an Iterative dynamic SSM Fusion (ISSF) framework consisting of  $N$  stacked DAFSSM modules, which integrate gating and modality alignment mechanisms to capture long-range spatial dependencies while mitigating feature misalignment.
- We propose an Adaptive feature Enhancement Downsampling (AEDown) layer to improve the robustness of multi-scale feature extraction and ensure information flow integrity, effectively preventing aliasing during high-frequency signal decimation.

- Extensive experiments demonstrate the effectiveness of our modules, achieving superior performance on the LLVIP [10], M3FD [11] and FLIR [12] datasets.

## 2. RELATED WORK

### 2.1. Multispectral Object Detection (MSOD)

A variety of approaches have been proposed for MSOD, advancing multimodal feature fusion through innovative techniques. Early methods predominantly relied on CNN-based architectures, employing simple concatenation or addition, which often limits the receptive field to local regions. To address this, CFT [6] introduces Transformers for mid-level feature fusion, and ICAFusion [12] employs dual cross-attention to capture complementary information. Despite their effective global modeling, Transformer-based solutions suffer from quadratic computational complexity. Recently, FMB [13] and SuperYOLO [14] have explored hidden state spaces and auxiliary tasks to balance efficiency and accuracy. However, these studies pay limited attention to the misalignment that arises after modality fusion, often neglecting critical local information.

### 2.2. Feature Alignment

Feature misalignment [15] after modality fusion remains a prevalent challenge [16, 17]. Researchers attribute this issue to asynchronous imaging across modalities, as well as scale and spatial inconsistencies in complex environments, alongside inference-level discrepancies. To address this, several strategies [18–20] have been explored, including convolutional offset prediction, constrained image matching, and multi-level feature alignment, all of which have achieved partial success. While existing alignment strategies mitigate spatial discrepancies, they largely rely on static receptive fields or fixed geometric constraints, proving inadequate for dynamic scenarios with non-rigid motion. Methods like convolutional offsets typically struggle with long-range dependencies, limiting efficacy across significant spatial gaps. Furthermore, treating alignment and fusion as disjoint stages often disrupts semantic coherence in feature representation. This underscores the necessity for a unified framework capable of dynamically modeling state transitions and aligning features in a context-aware manner, thereby addressing the intrinsic complexity of cross-modality fusion.

## 3. METHODS

### 3.1. Preliminaries

SSMs map an input sequence  $x(t)$  to an output sequence  $y(t)$  via hidden states  $h(t)$ . The update of the hidden state follows the following equation.

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad (1)$$

$$y(t) = \mathbf{C}h(t). \quad (2)$$

Under the Zero-Order Hold (ZOH) assumption, The continuous system's state-space matrices  $\mathbf{A}$  and  $\mathbf{B}$  are discretized into their discrete-time forms  $\mathbf{A}_d$  and  $\mathbf{B}_d$  using a positive real timescale  $\Delta$ .

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad (3)$$

$$\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(e^{\Delta\mathbf{A}} - \mathbf{I}) \cdot \Delta\mathbf{B}. \quad (4)$$

These transformations yield the discrete model, which can be expressed as:

$$h_t = \bar{\mathbf{A}} \cdot h_{t-1} + \bar{\mathbf{B}} \cdot x_t, \quad (5)$$

$$y_t = \mathbf{C} \cdot h_t, \quad (6)$$

$$\mathbf{K} = (\mathbf{C}\mathbf{B}, \mathbf{C}\mathbf{A}\mathbf{B}, \dots, \mathbf{C}\mathbf{A}^{L-1}\mathbf{B}), \quad (7)$$

$$y = x * \bar{\mathbf{K}}. \quad (8)$$

The hidden state  $h_t$  is updated based on  $h_{t-1}$  and the current input  $x_t$ , with the output  $y_t$  derived from  $h_t$  via a mapping matrix  $\mathbf{C}$ . The hidden state  $h_t$  is mapped back to the observation space through  $\mathbf{C}$ , yielding the output  $y_t$ . Here, the structured convolutional kernel is a matrix  $\bar{\mathbf{K}} \in \mathbb{R}^L$ , where  $L$  denotes the length of the input sequence. The system matrices are adjusted dynamically based on the current context of the input sequence.

### 3.2. Model Architecture

The model architecture is illustrated in Fig. 1. In the feature extraction stage, the backbone network employs a dual-stream structure. For single-stream processing of IR and RGB images, CSPDarkNet53 [21] is utilized to generate multi-scale features, denoted as  $\{C_1, C_2, C_3, C_4, C_5\}$ . During the single-modality feature extraction stage, the newly designed AEDown is employed for adaptive down-sampling, enhancing features within each modality. In the cross-modality feature extraction stage, at the feature maps  $\{C_3, C_4, C_5\}$ , an ISSF module fuses cross-modal features  $\bar{F}_{IR}$  and  $\bar{F}_R$ , dynamically aligning and generating complementary cross-modal information. This fused information is then passed to a multi-scale feature fusion network before being processed by the decoupled head.

### 3.3. Iterative dynamic State Space Fusion

As shown in Fig. 1, We propose the ISSF module, which contains  $N$  stacked **Dynamic Alignment Fusion SSM** that generate fused uni-modal representations. To strictly formulate our contribution, we define the Dynamic SSM as a **Time-Varying Linear Filtering** process. Unlike standard time-invariant models, we consider the discretized system matrices  $\bar{\mathbf{A}}$  and  $\bar{\mathbf{B}}$  as functions of the local input context. This effectively allows the model to adaptively adjust the receptive impulse response based on the local Signal-to-Noise Ratio (SNR), correcting phase divergence caused by motion or sensor misalignment. Their outputs are then subjected to a channel-wise concatenation and a subsequent convolution to efficiently learn a compact, unified inter-modal representation by leveraging the synergy across modalities.

The procedure, as shown in Fig. 1, illustrates how the DAF-SSM module exploits the combined advantages of the IR and RGB branches in multi-stage mid-level fusion, enhancing the global context modeling capability of SSM to reduce computational costs, while addressing local adaptive issues through content weighting. Initially, *LayerNorm* (denoted as  $f(\cdot)$ ) is applied to the input feature maps  $\bar{F}_R$  and  $\bar{F}_{IR}$ , yielding  $x^{l-2}$  and  $y^{l-2}$ , followed by dynamic offsets from deformable convolutions to assist feature alignment. The long-range dependency modeling of the SSM is then used to capture global relations. This process is represented by:

$$x^{l-2} = f(\bar{F}_R) \quad y^{l-2} = f(\bar{F}_{IR}) \quad (9)$$

$$x^{l-1} = g(x^{l-2}) \quad y^{l-1} = g(y^{l-2}) \quad (10)$$

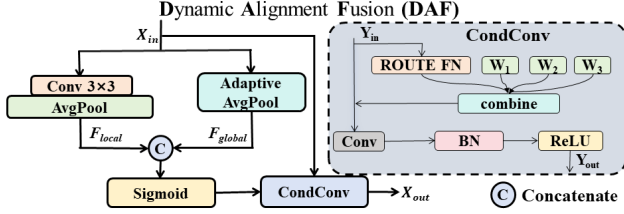
where  $f(\cdot)$  is *LayerNorm*, and  $g(\cdot)$  is *SSM(SiLU(DCNv2(\cdot)))*. Next, a gating mechanism is employed to modulate the extracted features, and dynamic alignment fusion is used to avoid misalignment of features after cross-modal fusion. The final output, complemented by residual connections, is expressed as:

$$x^l = x^{l-1} \odot x^{l-2} \quad y^l = y^{l-1} \odot y^{l-2} \quad (11)$$

$$\tilde{F}_R = h(x^l \oplus y^{l-1}) \oplus \bar{F}_R \quad (12)$$

$$\tilde{F}_{IR} = h(x^{l-1} \oplus y^l) \oplus \bar{F}_{IR} \quad (13)$$

where  $h(\cdot)$  denotes  $DAF(\cdot)$ , and  $\tilde{F}_R$ ,  $\tilde{F}_{IR}$  represent the output feature maps of the RGB and IR modalities, respectively.



**Fig. 2:** Illustration of DAF architecture. CondConv dynamically generates convolution kernels by learning a set of expert kernels and predicting weights for each input via a gating network. These weights combine the expert kernels into a dynamic kernel for the convolution operation, enabling adaptive feature extraction and improving efficiency.

Within the **Dynamic Alignment Fusion** module in Fig. 2, dynamic weight prediction based on global and local features is used to generate new content-optimized convolution kernels, addressing the feature misalignment issue post-modality fusion from a dynamic perspective. The design features a dual-branch structure: one branch extracts local features  $F_{local}$  via  $Conv_{3 \times 3}$  and  $AdaptiveAvgPool$ , while the other performs global pooling to obtain  $F_{global}$ . These features are then concatenated into a more comprehensive feature vector. CondConv utilizes the sigmoid function to generate dynamic weight distributions and determine the number of expert convolutions for dynamic prediction, ultimately generating convolution kernels that yield superior feature representations. The core computation of dynamic prediction is as follows:

$$W_{final} = \sum_{i=1}^{N_{experts}} S_i \cdot W_i \quad (14)$$

where  $W_{final}$  denotes the generated final kernel,  $N_{experts}$  represents the total number of expert convolution kernels,  $S_i$  and  $W_i$  denote the dynamic weight and the weight of the  $i$ -th expert kernel, respectively.

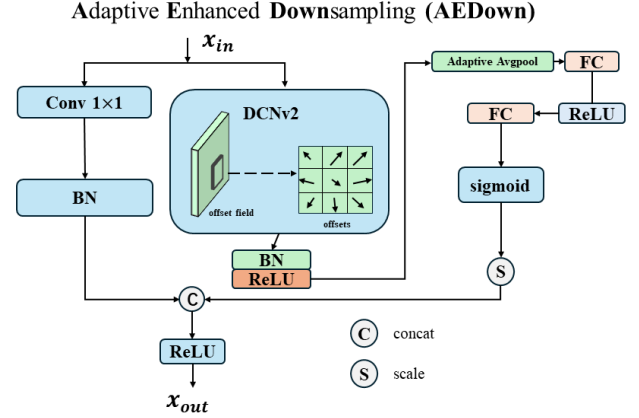
### 3.4. Adaptive Enhanced Downsampling

To mitigate spatial information loss during downsampling and strengthen the robustness of multi-scale feature extraction, we designed the **Adaptive Enhanced Downsampling** module from a dynamic perspective. As shown in Fig. 3, the extraction process is divided into three parts. First, in the dynamic branch, deformable convolution is used: based on the input, the sampling points of the convolutional kernel are dynamically offset to achieve shape adaptability, which can be represented by:

$$\{\Delta p, \Delta m\} = \sigma(f_{pred}(\mathbf{X})) \quad (15)$$

where  $\mathbf{X}$  is the input feature map,  $f_{pred}$  is the prediction convolution layer,  $\Delta p$  is the learned offset adjusting the kernel's sampling positions, and  $\Delta m$  is the modulation mask, generated by the Sigmoid function.

Then, after adaptive average pooling, two fully connected layers and their activation functions are applied to implement lightweight



**Fig. 3:** Illustration of AEDownsampling architecture. DCNv2 extends standard convolutions by learning dynamic offsets for sampling locations, allowing the model to adaptively focus on relevant regions of the input. These offsets are learned during training, enabling the network to handle spatial transformations like rotations and scale variations more effectively.

channel-level weighting and enhance salient features, with the weighting process represented by:

$$\mathbf{Y}_{out} = \mathbf{s} \otimes \mathbf{X} \quad (16)$$

where  $\mathbf{X}$  is the input feature map,  $\mathbf{s}$  is the channel weight vector, and  $\otimes$  represents element-wise multiplication to apply the weights.

Finally, a residual connection with the original branch preserves the information flow:  $f_{dynamic}(\cdot)$  is the output of the dynamic branch, and  $f_{static}(\cdot)$  is the output of the static branch.

$$\mathbf{X}_{out} = \text{ReLU}(f_{dynamic}(\mathbf{X}) + f_{static}(\mathbf{X})) \quad (17)$$

## 4. EXPERIMENTS AND ANALYSIS

### 4.1. Experimental Setup

All experiments are implemented based on CUDA12.1 torch 2.0.1 Python 3.8 and run on RTX 4090 GPU. The training process spans 150 epochs, with a batch size of 4. The image size is set to  $640 \times 640$ . The SGD optimizer is used with the initial learning rate of 0.01 and the momentum of 0.91.

### 4.2. Datasets

In this study, three widely recognized Multi-Spectral Object Detection (MSOD) datasets are adopted to evaluate the effectiveness of the proposed method: LLVIP, M3FD and FLIR.

LLVIP is a synchronized visible-infrared dataset collected in low-light environments, comprising 15,488 image pairs. It is designed to advance research in pedestrian detection and image fusion.

The M3FD dataset consists of images collected across diverse environments, including varying lighting, seasonal, and weather conditions. It annotates six categories, commonly encountered in autonomous driving scenarios.

The FLIR dataset consists of day and night scenes, encompassing five categories: people, car, bike, dog, and other vehicles. It is designed to facilitate the development of vision-based systems for low-light environments.

### 4.3. Experiments Results

**Table 1:** Comparison of Other Methods on LLVIP [10] Dataset.

| Method                     | Modality | Type        | Backbone      | mAP <sub>50</sub> (%) | mAP(%)      |
|----------------------------|----------|-------------|---------------|-----------------------|-------------|
| RTDETR-l [22]              | IR       | Transformer | HGNetV2 [22]  | 95.7                  | 62.8        |
| YOLOv8-l [21]              | IR       | One-stage   | CSPDarknet53  | 95.1                  | 61.9        |
| Faster R-CNN [23]          | IR       | Two-stage   | ResNet50 [1]  | 92.6                  | 50.7        |
| RTDETR-l                   | RGB      | Transformer | HGNetV2       | 88.4                  | 48.6        |
| YOLOv8-l                   | RGB      | One-stage   | CSPDarknet53  | 91.9                  | 53.4        |
| Faster R-CNN               | RGB      | Two-stage   | ResNet50      | 88.9                  | 47.6        |
| ProEn [24]                 | IR+RGB   | Two-stage   | ResNet50      | 93.4                  | 51.6        |
| CSAA [25]                  | IR+RGB   | Two-stage   | ResNet50      | 94.4                  | 59.2        |
| DIVFusion [26]             | IR+RGB   | One-stage   | YOLOv5 [21]   | 89.8                  | 52.1        |
| C <sup>2</sup> Former [27] | IR+RGB   | Transformer | VGG-16+ResNet | 95.5                  | 62.8        |
| CMDS(Ours)                 | IR+RGB   | One-stage   | CSPDarknet53  | <b>96.9</b>           | <b>64.4</b> |

**Table 2:** Comparison of Other Methods on M<sup>3</sup>FD [11] Dataset.

| Methods          | People      | Car         | Bus         | Motor       | Lamp        | Truck       | mAP <sub>50</sub> |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| ReC [28]         | 79.4        | 78.9        | 91.8        | 69.3        | 87.4        | 70.0        | 79.5              |
| TarDAL [11]      | 81.5        | <b>94.8</b> | 81.3        | 69.3        | 87.1        | 68.7        | 80.5              |
| DeF [29]         | 80.8        | 92.5        | 83.0        | 69.4        | <b>87.8</b> | 71.4        | 80.8              |
| SuperFusion [30] | 83.7        | 91.0        | 93.2        | 77.4        | 70.0        | <b>85.8</b> | 83.5              |
| CDDFuse [31]     | 81.6        | 92.5        | 82.6        | 71.6        | 86.9        | 71.5        | 81.1              |
| E2E-MFD [32]     | 80.6        | 91.5        | 90.4        | 79.9        | 87.1        | 84.2        | 85.7              |
| CMDS(Ours)       | <b>84.5</b> | 91.6        | <b>93.4</b> | <b>84.8</b> | 86.1        | 81.9        | <b>87.1</b>       |

The results on the LLVIP dataset are summarized in Table 1. YOLOv8-l and RTDETR-l achieve the best results under RGB and IR modalities respectively. Our proposed dual-stream detector with cross-modality dynamic SSM fusion (CMDS) attains the highest performance. Specifically, our CMDS outperforms the second-best method, C<sup>2</sup>Former, by 1.6% in mAP and 1.4% in mAP<sub>50</sub>, confirming the superiority of our dynamic alignment mechanism. Table 2 presents a comparison between the proposed CMDS and existing methods on the M<sup>3</sup>FD dataset. CMDS achieves the highest mAP<sub>50</sub> of 87.1%, outperforming the recent E2E-MFD (85.7%). CMDS achieves the highest average precision (AP) in the People, Bus, and Motor categories, with values of 84.5%, 93.4%, and 84.8%, respectively. Notably, in the Motor category, which is prone to misalignment during fusion due to the dynamic nature of motion, CMDS outperforms other methods with a clear advantage.

**Table 3:** Ablation Study with in Different Detectors on FLIR.

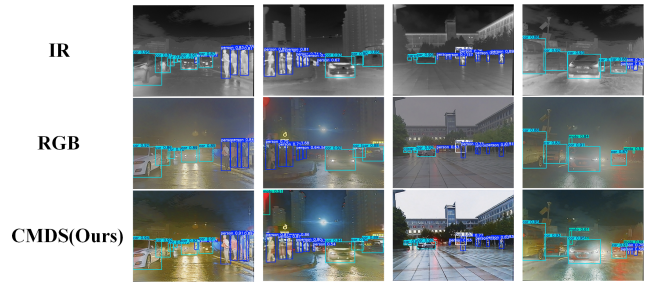
| Method             | Detector     | mAP <sub>50</sub> (%) | mAP(%) | Latency(ms) |
|--------------------|--------------|-----------------------|--------|-------------|
| Transformer-Fusion | Faster R-CNN | 77.3                  | 39.1   | 38.7        |
| CNN-Fusion         |              | 73.1                  | 37.1   | 35.6        |
| ISSF (Ours)        |              | 78.2                  | 40.4   | 36.9        |
| Transformer-Fusion | YOLOv8       | 79.3                  | 40.3   | 16.4        |
| CNN-Fusion         |              | 74.5                  | 37.2   | 14.5        |
| ISSF (Ours)        |              | 80.0                  | 41.2   | 15.6        |

To further validate the contributions of each component, we conduct ablation studies. As shown in Table 3, the proposed ISSF module demonstrates strong scalability: when applied to other detectors at different stages, it yields significant accuracy gains with only marginal increases in latency. Table 4 indicates that dynamic alignment and fusion of cross-modality information after SSM processing improves detection accuracy, while adaptive enhancement in the downsampling layers also contributes positively. The joint effect of these components significantly optimizes the detector’s performance in the context of MSOD.

**Table 4:** Ablation study with configurations on LLVIP [10] Dataset.

| Methods | ISSF |     | AEDown | mAP <sub>50</sub> (%) | mAP(%) |
|---------|------|-----|--------|-----------------------|--------|
|         | SSM  | DAF |        |                       |        |
| 1       | ✓    |     |        | 95.3                  | 62.5   |
| 2       |      | ✓   |        | 93.1                  | 59.7   |
| 3       |      |     | ✓      | 94.6                  | 61.4   |
| 4       | ✓    | ✓   |        | 96.1                  | 63.3   |
| 5       | ✓    |     | ✓      | 95.8                  | 63.1   |
| 6       | ✓    | ✓   | ✓      | 96.9                  | 64.4   |

Visualization results, as shown in Fig. 4, demonstrate that, compared to the individual visible light and infrared modalities, the fused CMDS not only exhibits higher confidence but also greater robustness, maintaining stable detection performance under diverse lighting and weather conditions.



**Fig. 4:** Comparison of IR, RGB, and CMDS inference results on M<sup>3</sup>FD [10] Dataset.

Fig. 5 illustrates a comparative analysis of images fused using different methods. The results demonstrate that the proposed CMDS method effectively addresses feature misalignment and information loss. Notably, no misalignment is observed in overlapping objects; the method ensures a strong focus on the target objects without surplus background information, thereby achieving cross-modal redundancy suppression.



**Fig. 5:** Comparison of the Original IR and RGB Images with the Cross-Modal Fused Images from Other Methods on the LLVIP [10] dataset.

## 5. CONCLUSION

This paper introduces a novel dual-stream detector fusion architecture based on State Space Models (SSM) from a dynamic perspective, designed to elevate feature representation across modalities while addressing challenges such as feature misalignment and the limitations of local receptive fields. By exploiting the synergy between SSM for global context capture and DAF for content-driven convolution kernel optimization, along with AEDown for dynamic enhancement of modality-specific salient features, the fusion process is significantly improved. The collaboration of these components not only improves detection accuracy but also reinforces model robustness, providing valuable insights for advancing future cross-modal fusion research.

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